MHF 4U1

CHAPTER 5: TRIGONOMETRIC FUNCTIONS

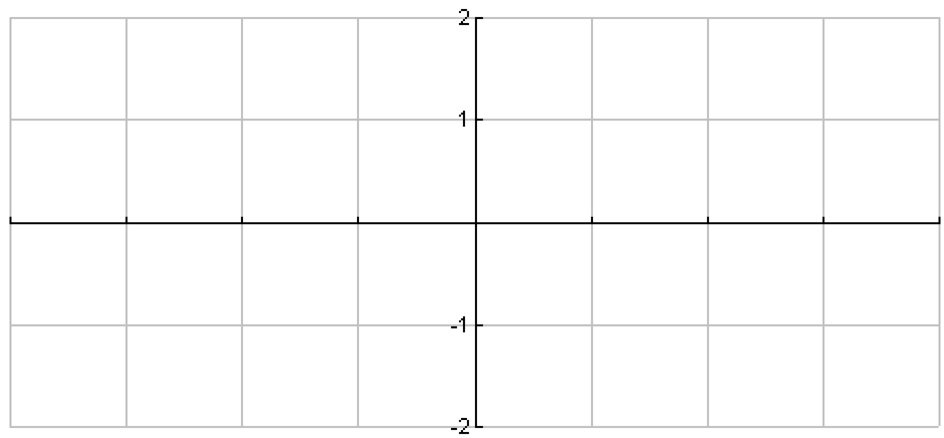
Day	Section	Homework
2	**5.1 Graphs of Sine, Cosine and Tangent Functions**	**Day 1: Page 258 # 1 – 11** **Day 2: Page 258 # 12-14, 17, 18, 21 (graph by hand)** **Page 269 #16 (omit iv)**
1.5	**5.3 Sinusoidal Functions of the form** ***f(x) =asin [k(x-d)] + c*** **and** ***f(x) =acos [k(x-d)] + c***	**Day 1: Page 275 # 1-3, 4(a-d), 5ab,6ab, 10a** **Day 2: Page277 #7,10b,11b,12,13a,14,15,19,20,23**
2	**5.2 Graphs of Reciprocal Trigonometric Functions**	**Page 267 # 1-7 (e.o), 10, 15, 18**
2	**5.4 Solve Trigonometric Equations**	**Page 287 # 1, 3, 5, 7, 9 –12** **Page 287 # 13 – 20, 25, 26, 28**
2	**REVIEW**	**Page 300 # 1 – 5, 7 - 12** **Page 302 # 1-15** **Handout**
1	**CHAPTER 5 TEST**	

CHAPTER 5 TEST DATE:

5.1 GRAPHS OF SINE AND COSINE FUNCTIONS

Given the relation **y = sin** θ, fill in the chart below using the unit circle. Graph the relation on the grid provided.

θ (degrees)	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
θ (radians)	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
sin θ	0								



Function? ____________________

Periodic? ____________________

Period in radian measure ____________________

Maximum value ____________________

Minimum value ____________________

Amplitude ____________________

y-intercept ____________________

Zeros $[0, 2\pi]$ ____________________

Domain ____________________

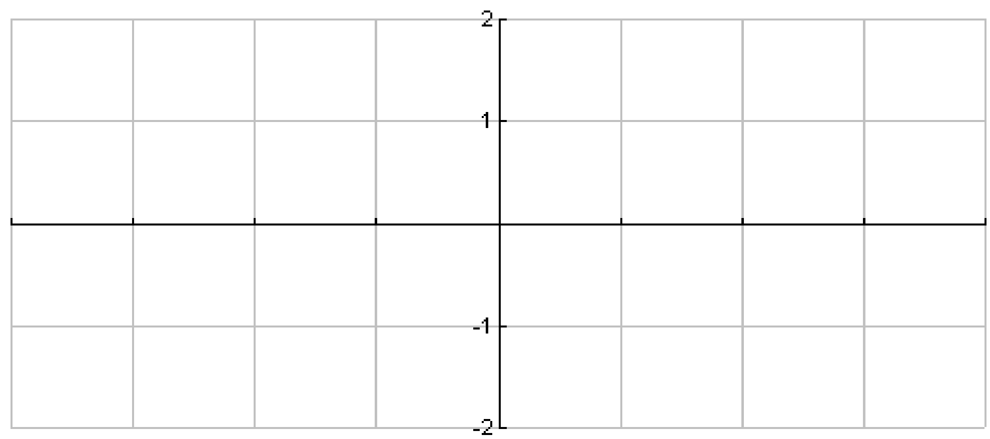
Range ____________________

> A function is periodic if it has a pattern of y-values that repeats at regular intervals.
>
> The horizontal length of one cycle (one complete pattern) is called the period of the function.

> The amplitude of a function is ½ the difference between the max and min values.

Given the relation **y = cos θ**, fill in the chart below using the unit circle. Graph the relation on the grid provided.

θ (degrees)	-360°	-270°	-180°	-90°	0°	90°	180°	270°	360°
θ (radians)	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
cos θ									



Function? ____________________

Periodic? ____________________

Period in radian measure ____________________

Maximum value ____________________

Minimum value ____________________

Amplitude ____________________

y-intercept ____________________

Zeros [0, 2π] ____________________

Domain ____________________

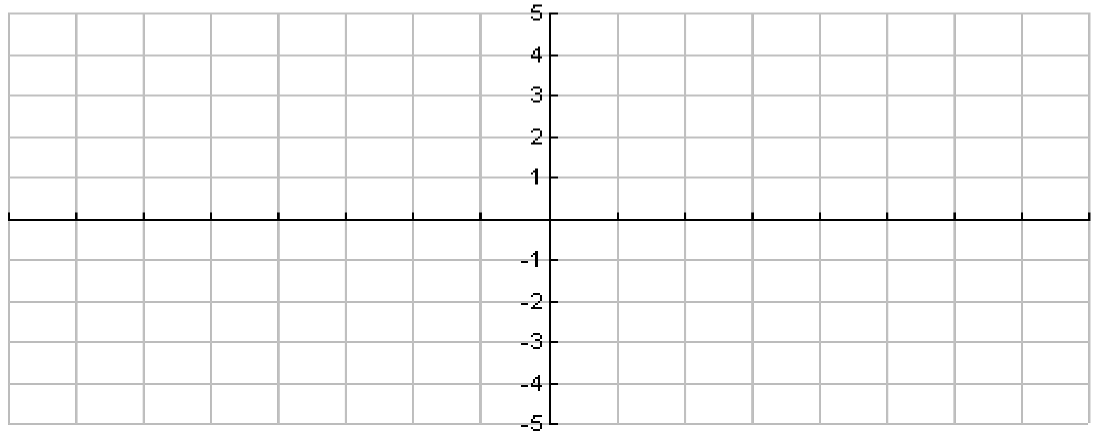
Range ____________________

Graphs of the form $y = \sin x + c$ and $y = \cos x + c$:

If $c > 0$, translate graph up c units.
If $c < 0$, translate graph down c units.

Mapping Rule: $(x, y) \rightarrow (x, y + c)$

Ex. 1: Transform the graph of $y = \sin x$ to sketch the graphs of $y = \sin x + 2$ and $y = \sin x - 3$.

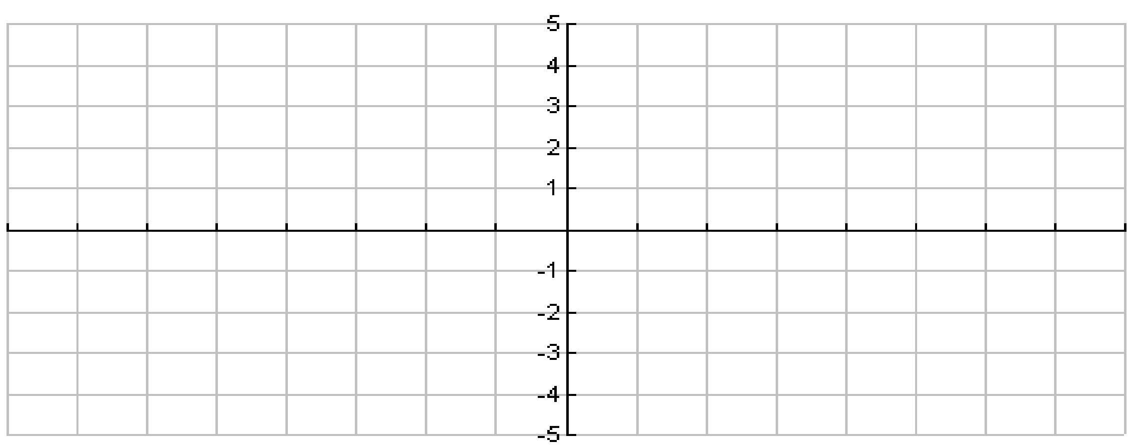


Graphs of the form $y = a \sin x$ and $y = a \cos x$:

If $a < 0$, graph is reflected in the x-axis.
If $a > 1$, graph is vertically stretched by a factor of a.
If $0 < a < 1$, graph is vertically compressed by a factor of a.

Mapping Rule: $(x, y) \rightarrow (x, ay)$

Ex. 2: Transform the graph of $y = \cos x$ to sketch the graphs of $y = \frac{1}{2}\cos x$ and $y = -3\cos x$.

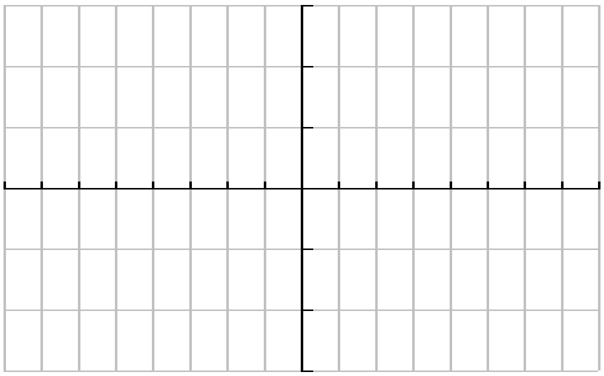


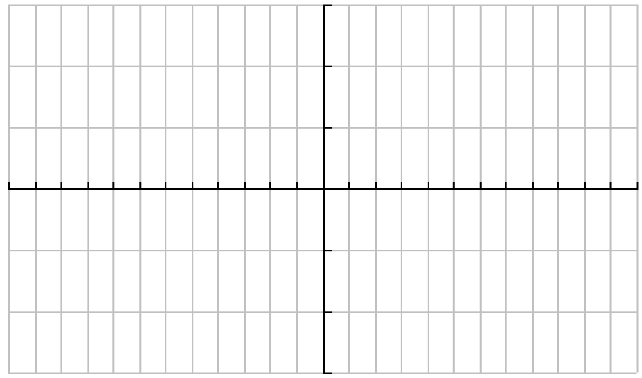
Graphs of the form y = sin(x - d) and y = cos(x - d):

If d > 0, graph is translated right d radians. Phase shift right d radians.
If d < 0, graph is translated left d radians. Phase shift left d radians.

Mapping Rule: (x, y) → (x + d, y)

Ex. 3: Transform the graph of $y = \sin x$ to sketch the graphs of $y = \sin(x + \frac{\pi}{4})$ and $y = \sin(x - \frac{\pi}{6})$.





Graphs of the form y = sin kx and y = cos kx:

If k< 0, graph is reflected in the y-axis.

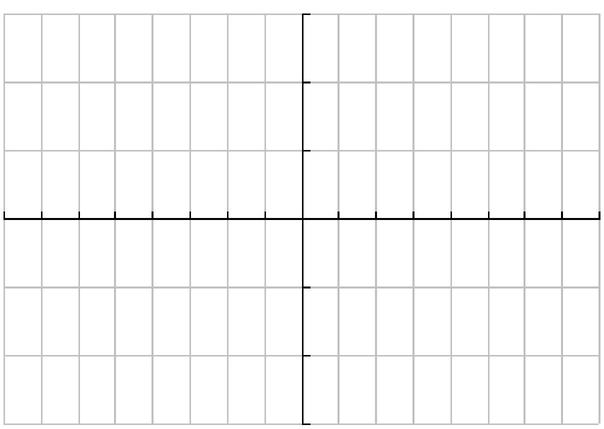
If k > 1, graph is compressed horizontally by a factor of $\frac{1}{k}$.

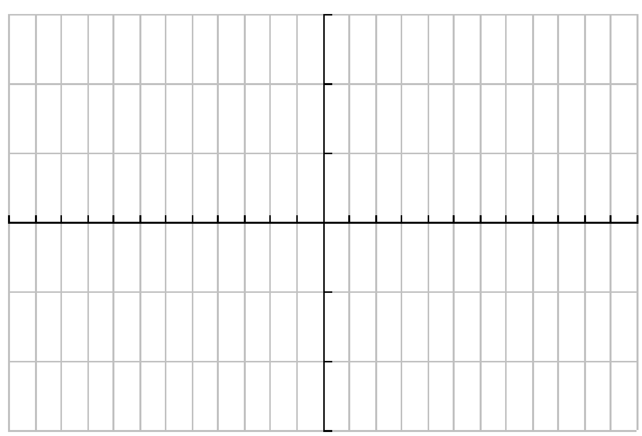
If 0 < k < 1, graph is stretched horizontally by a factor of $\frac{1}{k}$.

Period = $\frac{2\pi}{|k|}$

Mapping Rule: (x, y) → ($\frac{1}{k}$x, y)

Ex. 4: Transform the graph of $y = \cos x$ to sketch the graphs of $y = \cos 2x$ and $y = \cos\left(-\frac{1}{3}x\right)$.

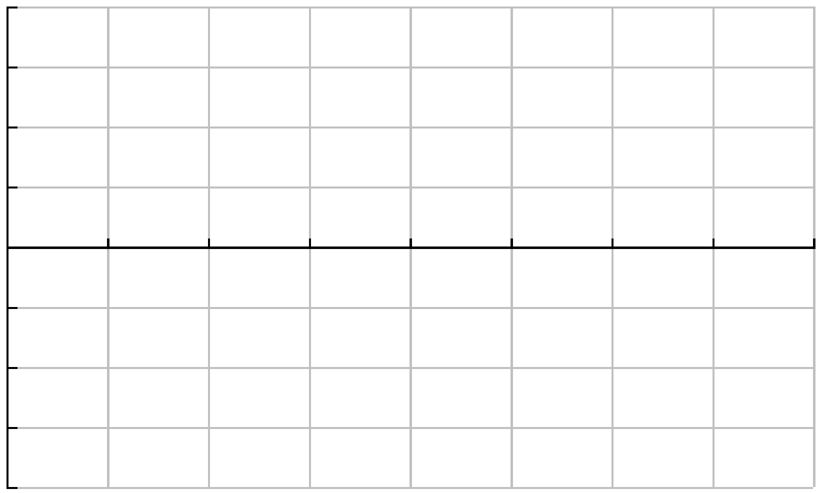




Try it now!

<u>INVESTIGATING THE GRAPH OF y = tanx:</u>

1. **Graph the functions of y = sinx and y = cosx on the same set of axes using a different colour for each.**



2. **Complete the table below using the 5 major points plotted from each function beside and the special triangle containing $\frac{\pi}{4}$.**

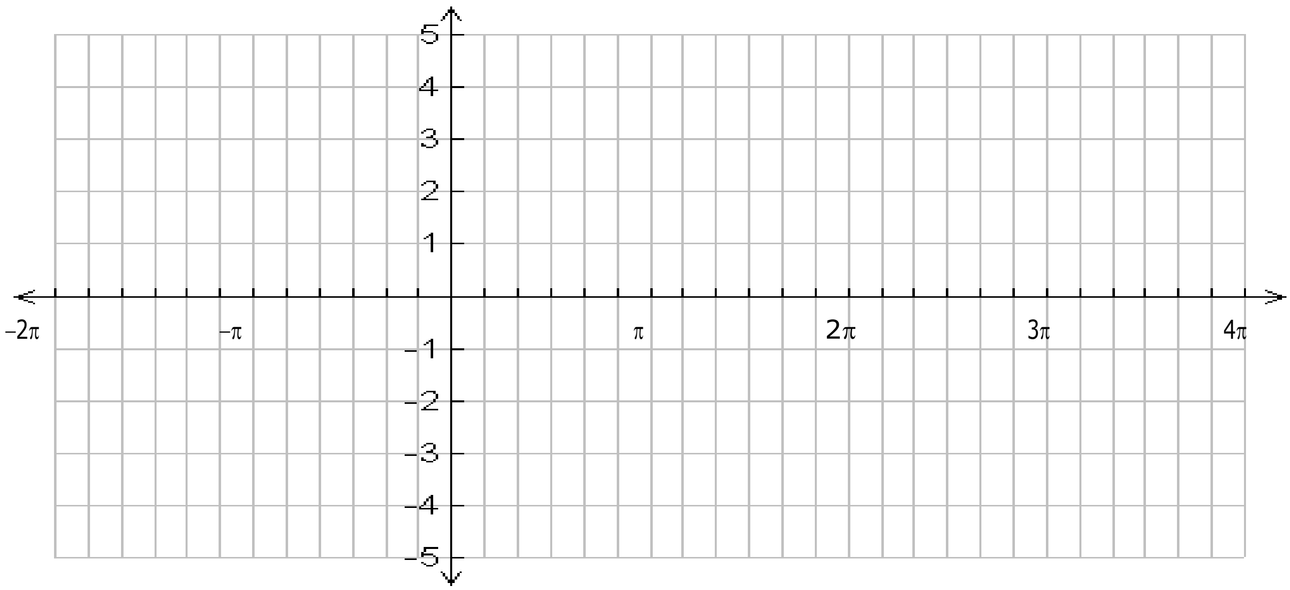
ANGLE (x)	y= sinx	y=cosx	y=tanx $= \frac{\sin x}{\cos x}$
0			
$\frac{\pi}{4}$			
$\frac{\pi}{2}$			
$\frac{3\pi}{4}$			
π			
$\frac{5\pi}{4}$			
$\frac{3\pi}{2}$			

3. **NOW GRAPH y=tanx using the table from #2 on the grid from #1**

5.1 DAY 2: GRAPH OF THE TANGENT FUNCTION AND APPLICATIONS OF SINE AND COSINE

Given the relation **y = tan θ**, fill in the chart below. Calculate the decimal value for tan θ rounded to the nearest tenth. Graph the relation on the grid provided. Extend the graph so that $-2\pi \le \theta \le 4\pi$.

Value of θ (degrees)	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
Value of θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
Decimal value of **tan θ**													



Function? ______________________

Periodic? ______________________

Period in radian measure ______________________

Maximum value ______________________

Minimum value ______________________

Amplitude ______________________

y-intercept ______________________

Zeros [0, 2π] ______________________

Domain ______________________

Range ______________________

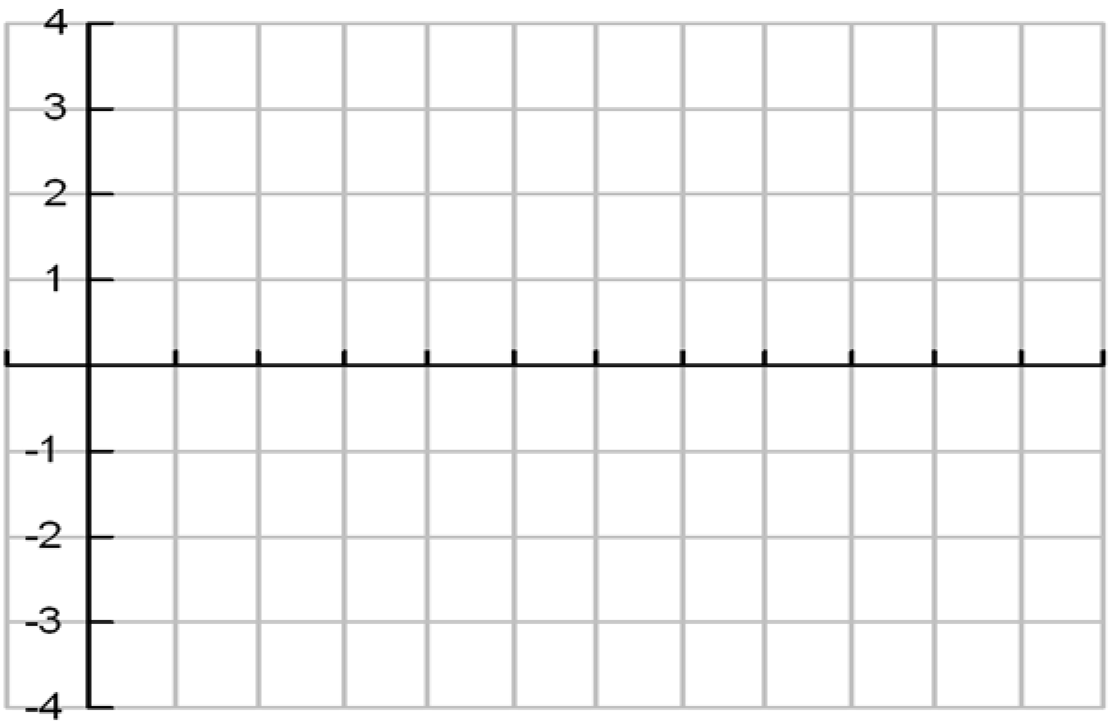
Vertical asymptotes ______________________

Example #1

A boat is bobbing up and down on the water. The distance between the boat's highest and lowest point is 4m. The boat moves from its highest to its lowest point and back to its highest point every 30 seconds.

a) Write a **cosine function** that models the movement of the boat in relation to the equilibrium.

b) Sketch three cycles of the graph on the grid below.



5.3 DAY 1: SINUSOIDAL FUNCTIONS OF THE FORM **f(x) = a cos/sin[k(x-d)] +c**

Ex 1: Consider the function: $y = 4\sin\left(2x + \frac{4\pi}{3}\right) - 5$

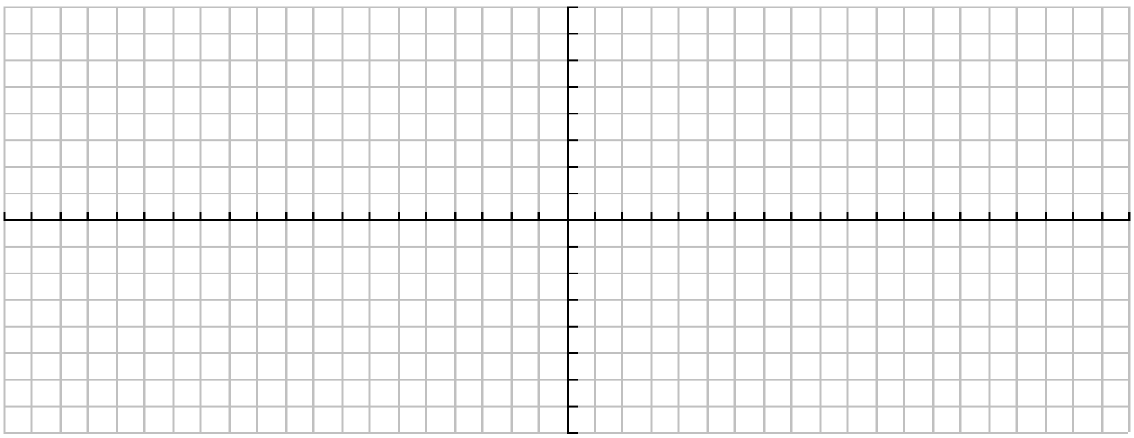
a) What is the amplitude? ____________

b) What is the period? _______________

c) Describe the phase shift.

d) Describe the vertical translation.

e) Graph & label the function, for 2 complete cycles.



f) Find the x-intercepts of the function.

g) State the domain and range of the two cycles.

D = ___________________________________

R = ___________________________________

Ex 2: Consider the function $y = -2\cos\left(\frac{1}{2}x - \frac{\pi}{12}\right) + 1.$

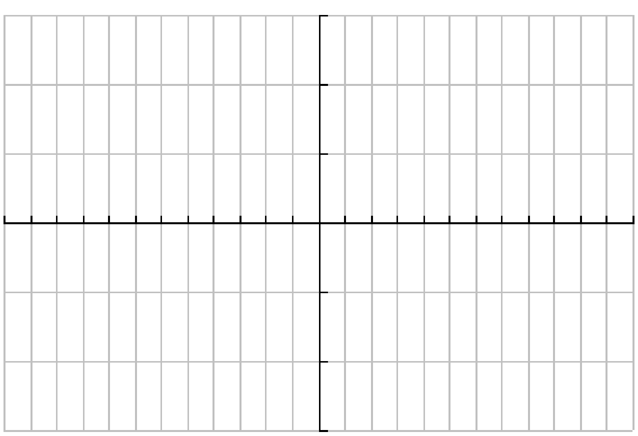
a) What is the amplitude?

b) What is the period?

c) Describe the phase shift.

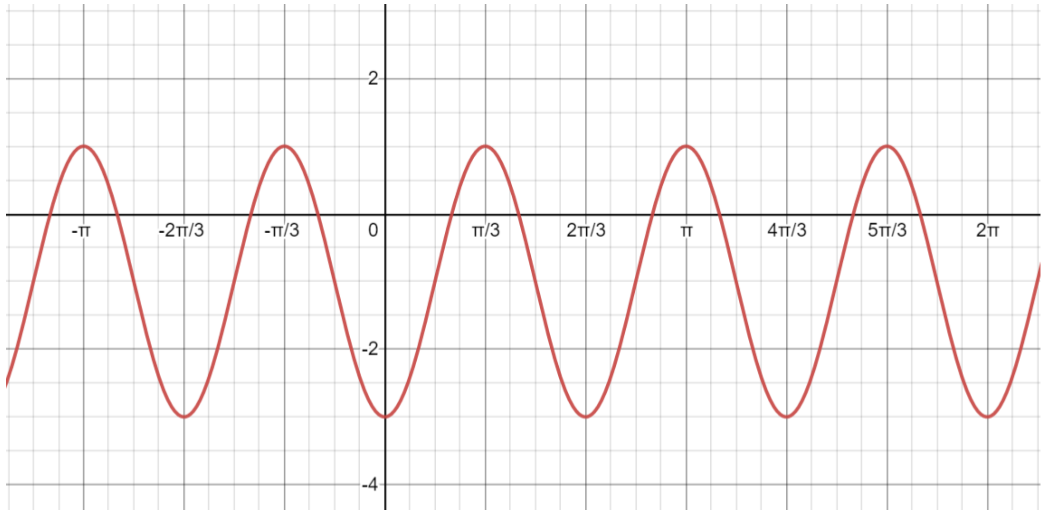
d) Describe the vertical translation.

e) Graph and label the function



f) State the domain and range.

<u>Ex 3</u>: Given the graph:



a) What is the maximum value? _______ What is the minimum value? _______

b) What is the amplitude?

c) What is the vertical translation?

d) What is the period?

e) What value of k will result in the required period?

f) Explain why the phase shift can have more than one value.

g) Suggest at least three values for the phase shift. Write possible sine and cosine equations for the above graph.

Try it now!

1. Given an equation y = 35sin[-6x-10π] -65. Identify the following:

a) amplitude = _________

b) period= _________

c) phase shift= _________

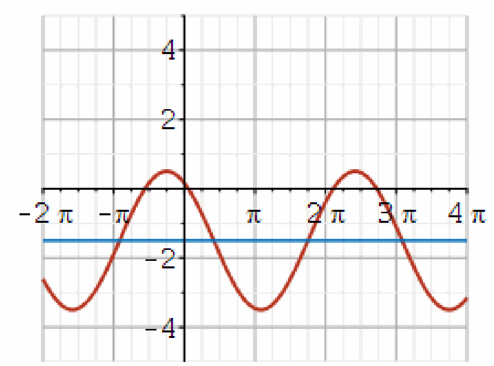
d) vertical shift= _________

e) minimum value= ________

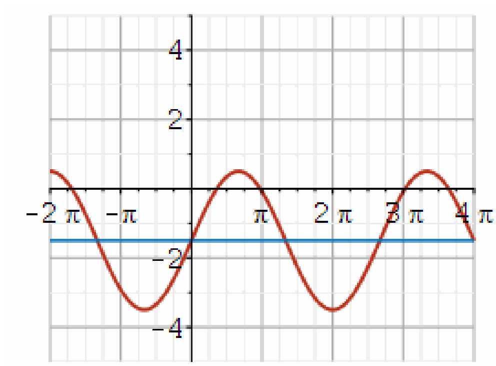
f) maximum value=________

2. Given an equation $y = 2\sin\left(\frac{3}{4}\left(x-\frac{7\pi}{4}\right)\right)+\frac{3}{2}$, determine which graph best represents the function:

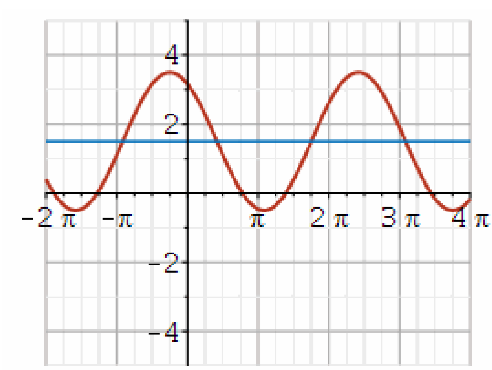
Graph A



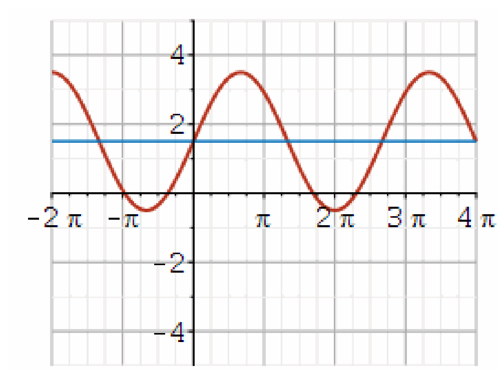
Graph B



Graph C



Graph D

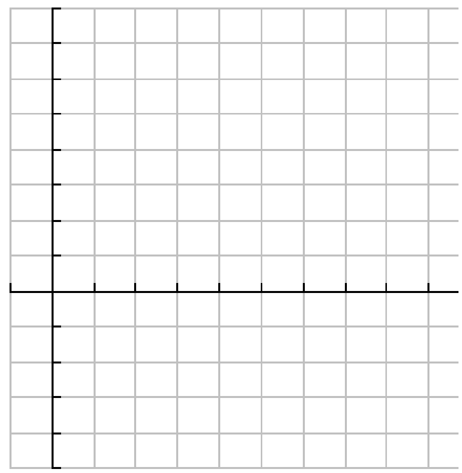


Try it now!

1. Write an equation for the sine graph with amplitude $\frac{1}{4}$, reflection in the x-axis, phase shift of $\frac{2\pi}{3}$, period of $\frac{5\pi}{4}$ and a vertical shift 5 down. ____

2. When seated, an average adult breathes in and out every 4 sec. The average minimum air in the lungs is 0.08 L, and the average maximum 0.82 L. If lungs have a minimum amount of air at t=0.

a) Write a cosine function to model the air in the lungs.



b) Graph

c) Determine the amount of air in lungs at 5.5 sec.

3. For $y = -\frac{1}{2}\sin\left[3x + \frac{\pi}{2}\right] - 5$

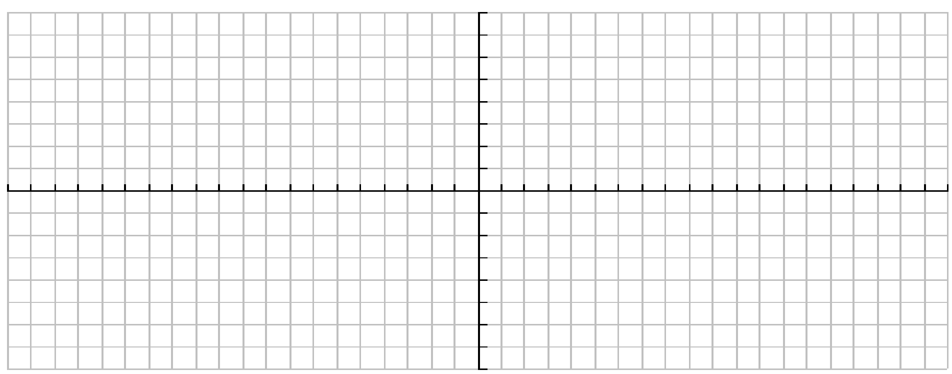
a) State the amplitude, period, phase shift and vertical translation

a = ___________

period = _________

PS = _____________

VT = _____________

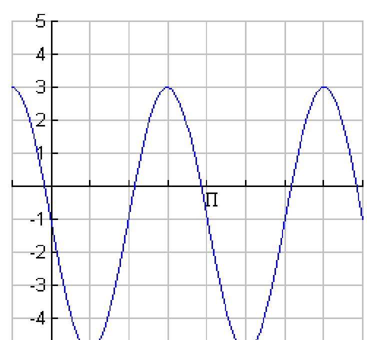


b) Sketch 2 cycles.

c) State the domain and range of your graph.

D =______________________ R=___________________________

4. Determine the equation of the graph:

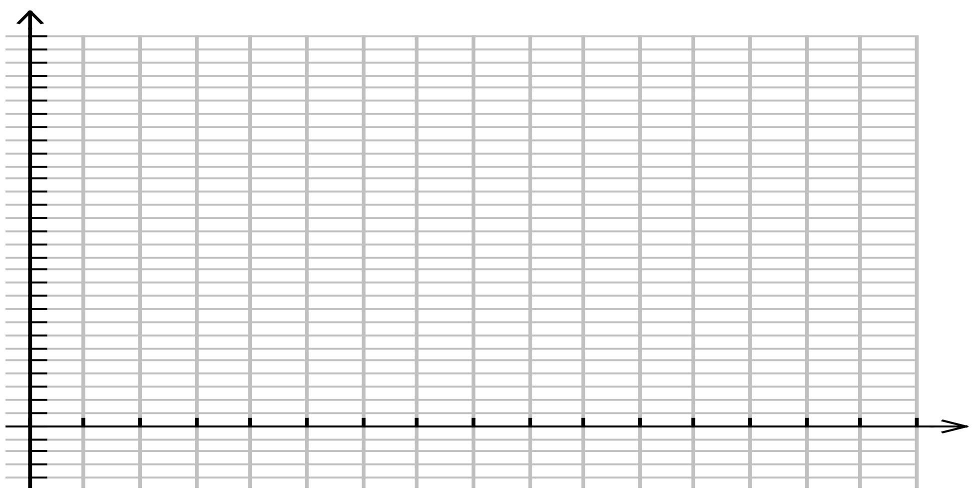


5.3 DAY 2: SINUSOIDAL FUNCTIONS OF THE FORM **$f(x) = a \cos/\sin[k(x-d)] + c$**

<u>Ex#1</u>:
A Ferris wheel at an amusement park completes one revolution every 40 seconds. The wheel has a diameter of 16 metres and its centre is 12 metres above the ground.

a) Starting at the lowest point, model the height, *h*, in metres above the ground, using a sine function of the form $h = a\sin[k(t-d)]+c$, where t represents time in seconds.

b) Sketch a graph of the model over 2 cycles.

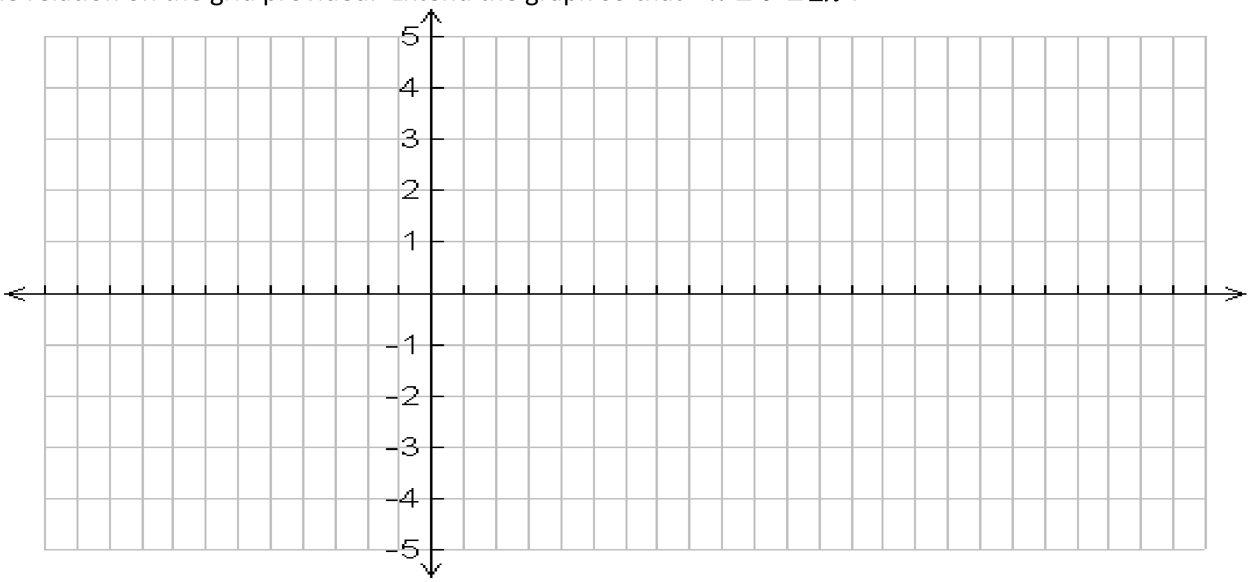


5.2 GRAPHS OF THE RECIPROCAL TRIGONOMETRIC FUNCTIONS

Given the relation **y = csc** $\boldsymbol{\theta}$*,* fill in the chart below. Calculate the decimal value for csc θ *(using sin* θ*)* rounded to the nearest hundredth.

$\boldsymbol{\theta}$	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
csc$\boldsymbol{\theta}$													

Graph the relation on the grid provided. Extend the graph so that $-\pi \le \theta \le 2\pi$.



Function? ____

Periodic? _______________ Period in radian measure _____________________

Equations of Asymptotes ____

Domain ____

Range ____

What happens when csc x = ± 1? _______________________________________

How do the positive/negative intervals compare to the graph of y = sin x?

____

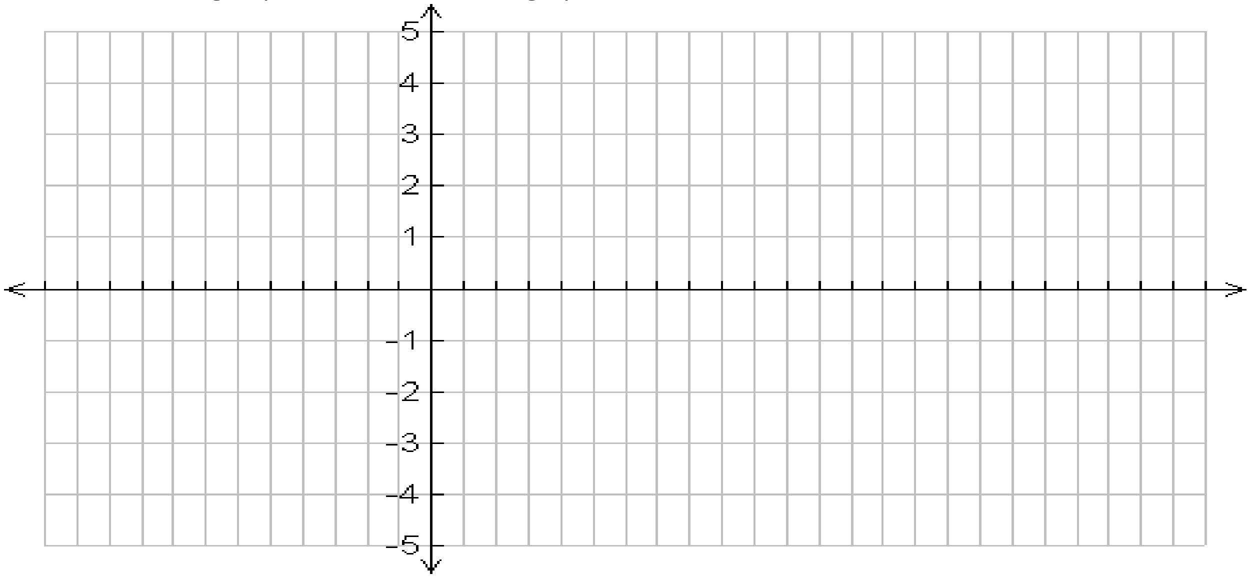
How do the intervals of increase/decrease compare to the graph of y = sin x?

____

Given the relation **y = sec** θ, fill in the chart below. Calculate the decimal value for sec θ *(using cos* θ*)* rounded to the nearest hundredth.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
secθ													

Graph the relation on the grid provided. Extend the graph so that $-\pi \le \theta \le 2\pi$.



Function? ____

Periodic? _______________ Period in radian measure ______________________

Equations of Asymptotes ____

Domain ____

Range ____

What happens when sec x = ± 1? ____

How do the positive/negative intervals compare to the graph of y = cos x?

____

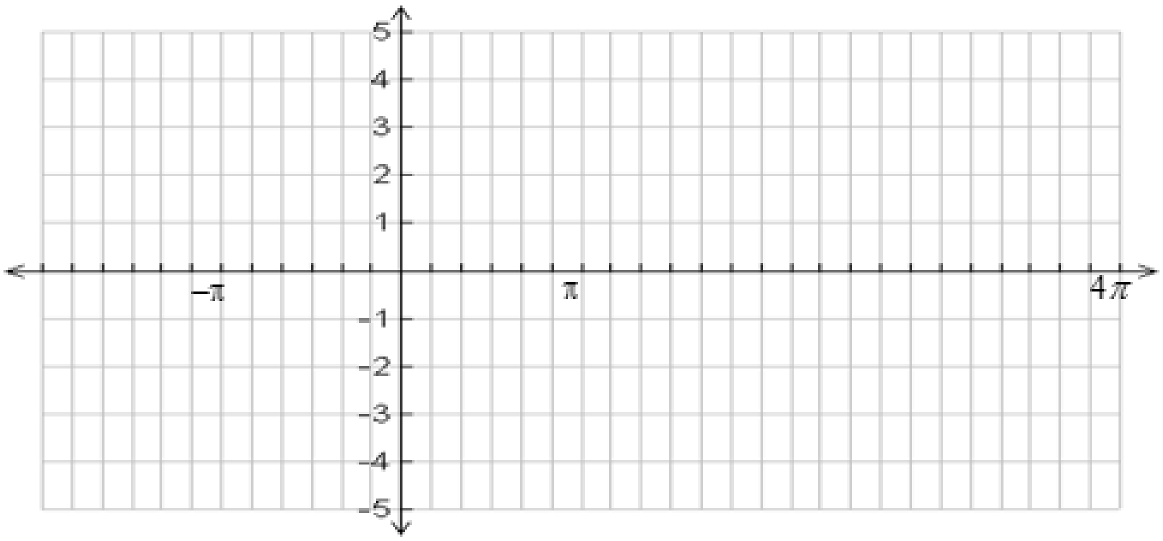
How do the intervals of increase/decrease compare to the graph of y = cos x?

____

Given the relation **y = cot** $\boldsymbol{\theta}$, fill in the chart below. Calculate the decimal value for cot θ *(using tan* θ*)* rounded to the nearest hundredth.

$\boldsymbol{\theta}$	0	$\frac{\pi}{6}$		$\frac{\pi}{3}$	$\frac{\pi}{2}$		$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
cot$\boldsymbol{\theta}$															

Graph the relation on the grid provided. Extend the graph so that $-2\pi \le \theta \le 4\pi$.



Function? ____

Periodic? _______________ Period in radian measure ______________________

Equations of Asymptotes ____

Domain ____

Range ____

Zeros $(-2\pi \le \theta \le 4\pi)$ ____

What happens when cot x = ± 1? ____

How do the positive/negative intervals compare to the graph of y = tan x?

____

How do the intervals of increase/decrease compare to the graph of y = tan x?

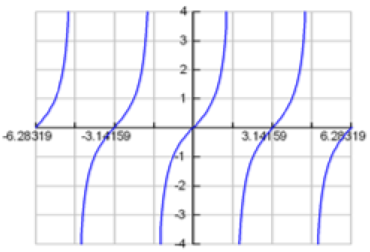
____

Ex. 1: Reciprocal and Inverse Notation

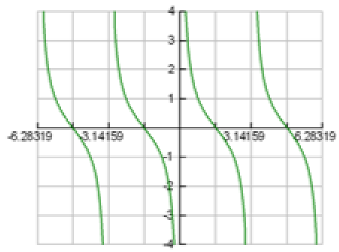
a) Explain the difference between $\sec\left(\frac{1}{2}\right)$ and $\cos^{-1}\left(\frac{1}{2}\right)$.

b) Use a calculator to illustrate the difference between the two expressions.

Ex. 2: Two successive transformations can be applied to the graph of y = tan x to obtain the graph of y = cot x. Describe one possible way to apply these compound transformations (many exist). Write the equation.

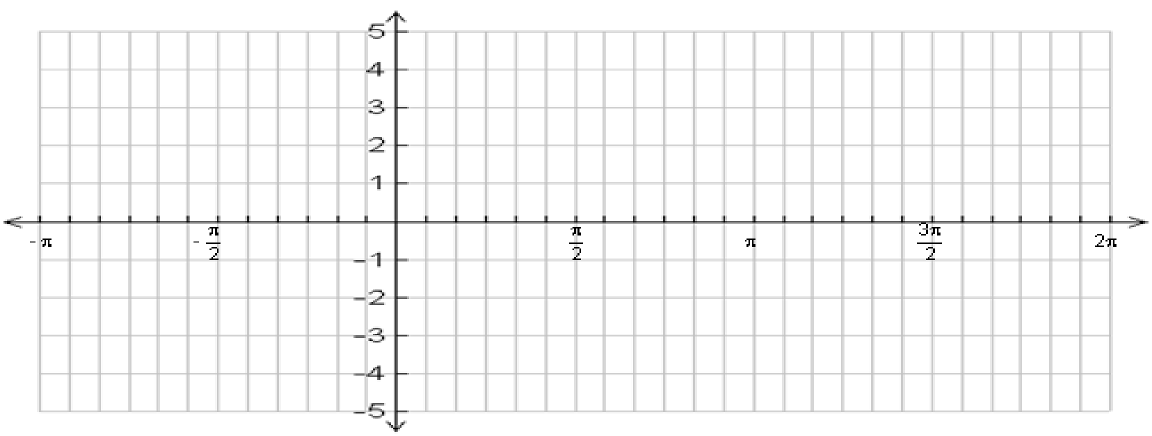


$y = tan\ x$

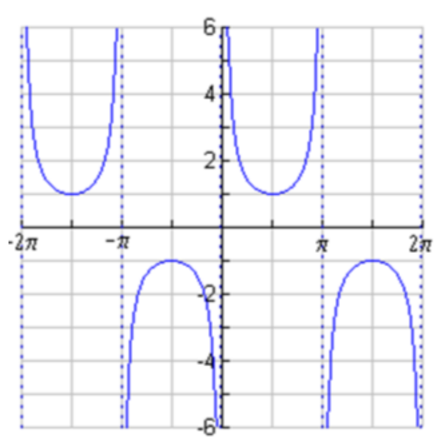


$y = cot\ x$

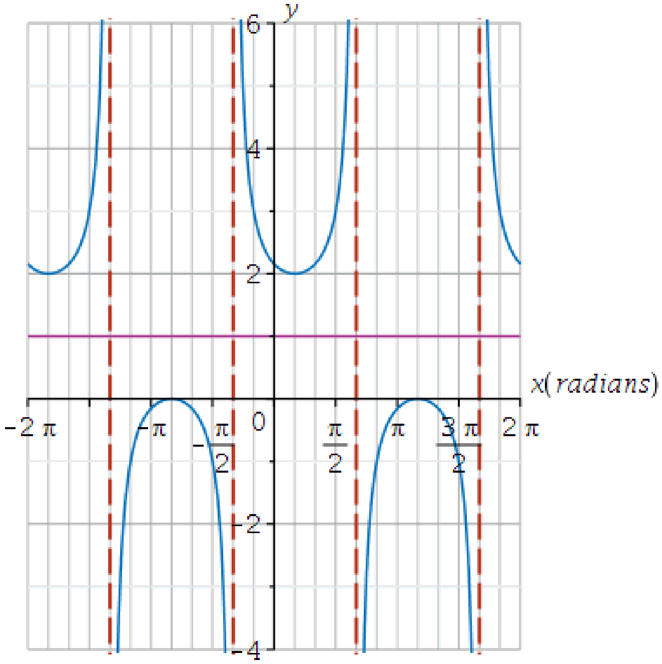
Ex. 3: Sketch the graph of y = sec 3x – 1 using transformations.



Ex. 4: Use the graph of y = csc x to solve for all values of x in the interval [0, 2 π] such that csc x = 5. (Hint: recall the CAST rule).



Try it now!



1. Given the following graph.

a) State the range:

____

b) State the x-values of the visible vertical asymptotes:

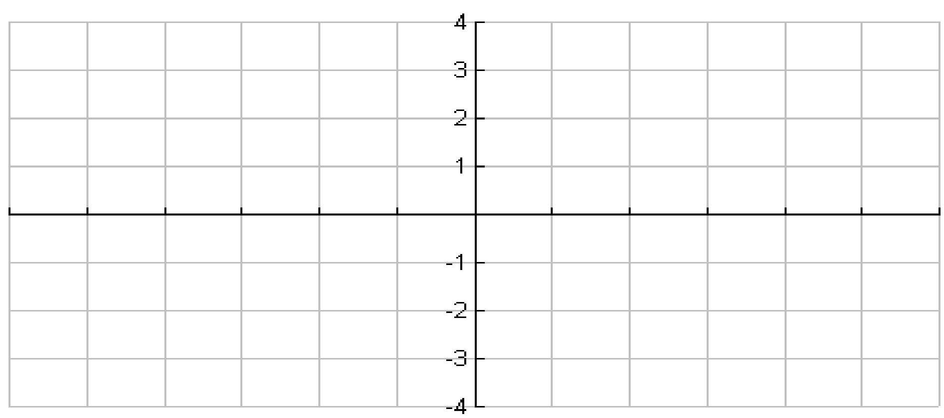
____

c) State the period: ________________

d) State a possible equation to represent the graph:

____

2. a) Sketch the graph of the function, y = csc x, on the domain $-3\pi \le x \le 3\pi$.
Completely label the graph.



b) Use your graph to determine the **number of solutions** to csc x = -2. **Show on your graph**.

c) Give the exact value of **ONE solution** to csc x = -2.

Try it now!

A rider’s height h, in m, above the ground at time t min, t≥0, can be modeled with the equation,

$$h = 30\sin\left(\frac{6\pi}{5}t\right) + 38$$

a) At what height does the rider get on the ride? ________________

b) What is the maximum height above the ground that a rider reaches on the ride? ______________

c) How many times in the first 3 minutes is the rider at the max height?

d) At what time, to the nearest second, is the rider 47 m above the ground for the first time during the ride?

Try it now!

Today, the high tide in Matthews Cove, New Brunswick, is at midnight. The water level at high tide is 7.5 m. The depth, d metres, of the water in the cove at time t hours is modeled by the equation $d(t) = 4 + 3.5\cos\frac{\pi}{6}t$.
Jenny is planning a day trip to the cove tomorrow, but the water needs to be at least 2 m deep for her to manoeuvre her sailboat safely.
How can Jenny determine the times when it will be safe for her to sail into Matthews Cove?

5.4 SOLVE TRIGONOMETRIC EQUATIONS

*Solving trigonometric equations means finding the zeroes of the corresponding graph.

Ex#1 Solve $2\sin x + 1 = 0$, on the interval x ε $[0,2\pi]$. Use exact values in radians.

Ex#2 Solve $\sin^2x - \sin x = 0$, on the interval x ε $[0,2\pi]$. Use exact values in radians.

Ex#3 Solve $3\cos^2x + 2\cos x - 1 = 0$ on the interval x ε $[0,2\pi]$ (to one decimal place).

Ex#4 Solve $2\csc^2x + \csc x - 1 = 0$ on the interval x ε $[0,2\pi]$ (to one decimal place).

5.4 DAY 2 : SOLVE TRIGONOMETRIC EQUATIONS

$\sin^2 x + \cos^2 x = 1$	**$\sin 2x = 2\sin x\cos x$**	**$\cos 2x = \cos^2 x - \sin^2 x$**
$1 + \tan^2 x = \sec^2 x$	**$\sin(x \pm y) = \sin x\cos y + \cos x\sin y$**	**$\cos(x\pm y) = \cos x\cos y - \sin x\sin y$**

Ex 1: Solve for $x\ \varepsilon\ [-\pi, 2\pi]$: $\sin 2x + \cos x = 0$ (exact solution).

Ex#2 Solve $2\sec^2 x - 3 + \tan x = 0$, on the interval $x\ \varepsilon\ [0,2\pi]$. (Use exact radians if possible.)

Ex 3: Solve $\cos 2x - 0.7 = 0$, for $0 \le x \le 2\pi$ (round to 2 decimal places).

Ex 4: Solve $\sin x\cos x + \cos x\sin x = 0.45$, for $0 \le x \le 2\pi$ (round to 2 decimal places).

1. Sketch a graph including 2 cycles of each of the following functions. State the domain and range for each.

a) $y = \sin 2x + 3$

b) $y = 2\cos\left(t + \frac{\pi}{3}\right) - 4$

c) $y = -\frac{1}{2}\sin\left(\frac{\pi}{2} - x\right)$

d) $y = -2\tan 4x + 1$

e) $y = -3\cos(6x - \pi) - 2$

f) $y = -2\sec 2x - 1$

g) $y = \frac{1}{2}\csc\left(\frac{1}{3}x - \frac{\pi}{3}\right)$

h) $y = \cot\left(\frac{1}{4}\vartheta + \frac{\pi}{2}\right)$

i) $y = \sec \pi(x + 3) + 2$

j) $y = 2\csc\frac{\pi}{2}\left(x + \frac{1}{4}\right)$

2. Solve each of the following for $x \in [0, 2\pi]$.

a) $\cos x = -\sqrt{0.5}$

b) $\sin^2\vartheta - 1 = 0$

c) $\cos 2\vartheta = 0$

d) $2\sin^2 x - 1 = \sin x$

e) $\sec x - 5 = 0$

f) $6\cos^2 x + \cos x - 1 = 0$

g) $8\cos^2 x + 14\cos x = -3$

h) $\csc x - 3\csc x \sec x = 0$

i) $\tan^2 x + 2\sec x - 7 = 0$

j) $2\tan^2\vartheta - \sqrt{12} = 0$

k) $2\csc^2 x - 8 = 0$

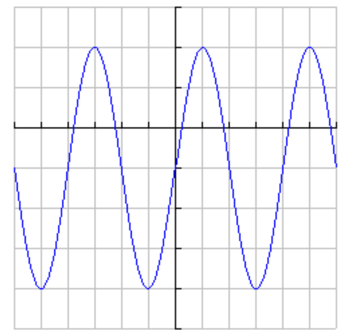
l) $\cos^2 x - \frac{64}{144} = 0$

m) $\sec 2x - 3 = 0$

n) $5\sin 3x + 2 = 0$

CHAPTER 5 REVIEW: MULTIPLE CHOICE QUESTIONS

1. What is the equation of this transformation of the graph of y =sinx?



a. y=3sin(2(x+1))

c. y = 3sin ($\frac{1}{2}x$) -1

b. y = 3sin(2x) -1

d. y = 2sin(3x) -1

2. One blade of a wind turbine is at counterclockwise angle of $\frac{\pi}{4}$ to the upward vertical at t=0, and rotates counterclockwise with one revolution every 2 h. The tip of the blade varies between 5 m and 41 m above the ground. Which equation is a model for the height of the blade tip?

a. h = 18cos (πt + $\frac{\pi}{4}$) + 23

c. h = 18cos(πt -$\frac{\pi}{4}$) – 23

b. h = 41cos(2(t+$\frac{\pi}{4}$)) - 5

d. h= 41cos(π(t+$\frac{\pi}{4}$)) – 36

3. Which set of value(s), in radians, is the solution of 5+7sinx = 0, where $-\pi \le x \le \pi$?

a. x = -0.80

c. x = 0.80, 2.35

b. x = -0.80,-2.35

d. x = -0.80, 0.80

4. The height of the tip of one blade of a wind turbine above the ground, h(t), can be modeled by h = 18cos (πt + $\frac{\pi}{4}$) + 23, where t is the time passed in seconds. Which time interval describes a period when the blade tip is at least 30 m above the ground?

a. $5.24 \le t \le 7.33$

c. $1.37 \le t \le 2.12$

b. $0.42 \le t \le 1.08$

d. $0.08 \le t \le 1.42$

5. Which set of values is the solution of (2sinx+1)(cosx-1)=0, where $0 \le x \le 2\pi$?

a. x = $\pi, \frac{7\pi}{6}, \frac{11\pi}{6}$

c. x = 0, $\frac{5\pi}{6}, \frac{7\pi}{6}, 2\pi$

b. x = $\frac{\pi}{6}, \pi, \frac{5\pi}{6}$

d. x = 0, $\frac{7\pi}{6}, \frac{11\pi}{6}, 2\pi$

6. The equation 2cos^{2}x + dcosx + e = 0 has solutions x = 0, $\frac{\pi}{3}, \frac{5\pi}{3}, 2\pi$ in the interval $0 \le x \le 2\pi$. What are the values of d and e?

a. d =-3, e = 2

c. d=1, e =3

b. d = 2, e= 3

d. d = -3, e = 1